INCH-POUND
MIL-DTL-16377/10B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/10A(SH)
29 July 1981

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SOLID-STATE EQUIVALENT, GENERAL LIGHTING, DIRECTIONAL, 32 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 145.4, 145.5, 145.4SSL, AND 145.5SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 6 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

AMSC N/A FSC 6210

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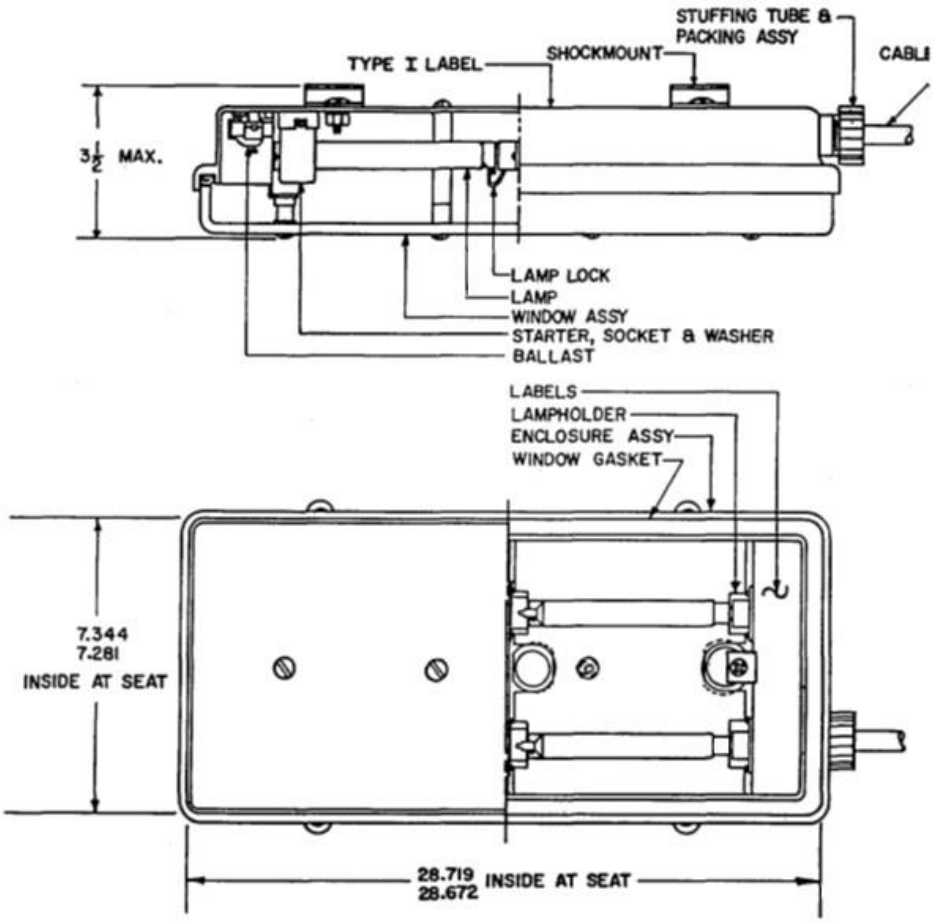


FIGURE 1. Typical view of symbol 145.4.

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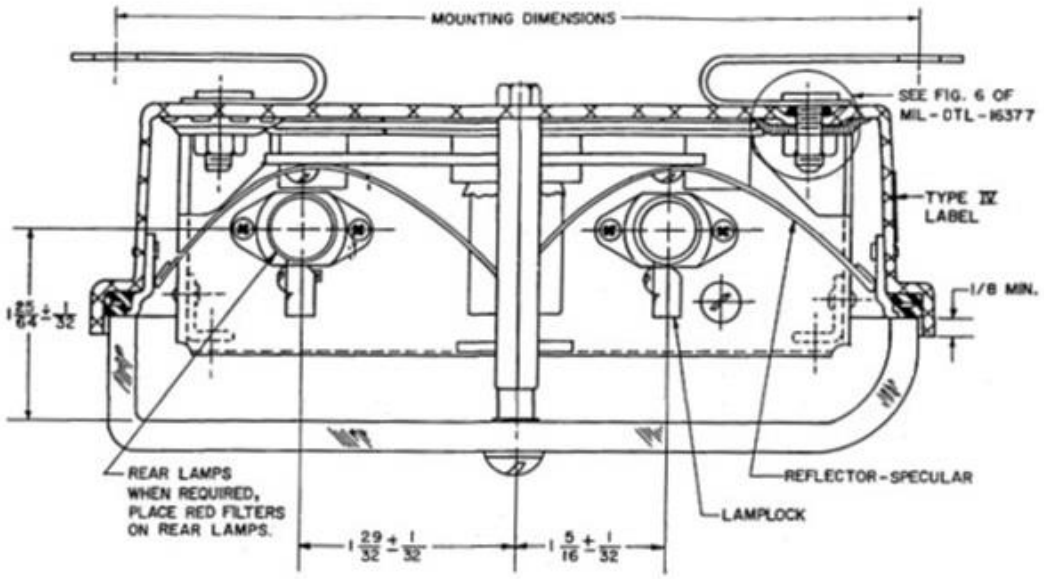


FIGURE 2. Typical section of symbol 145.4.

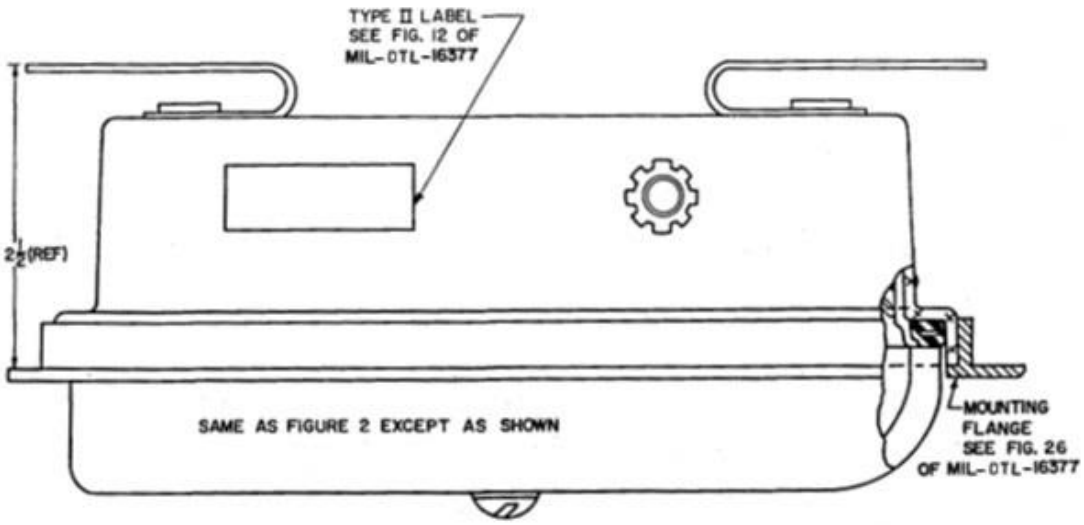
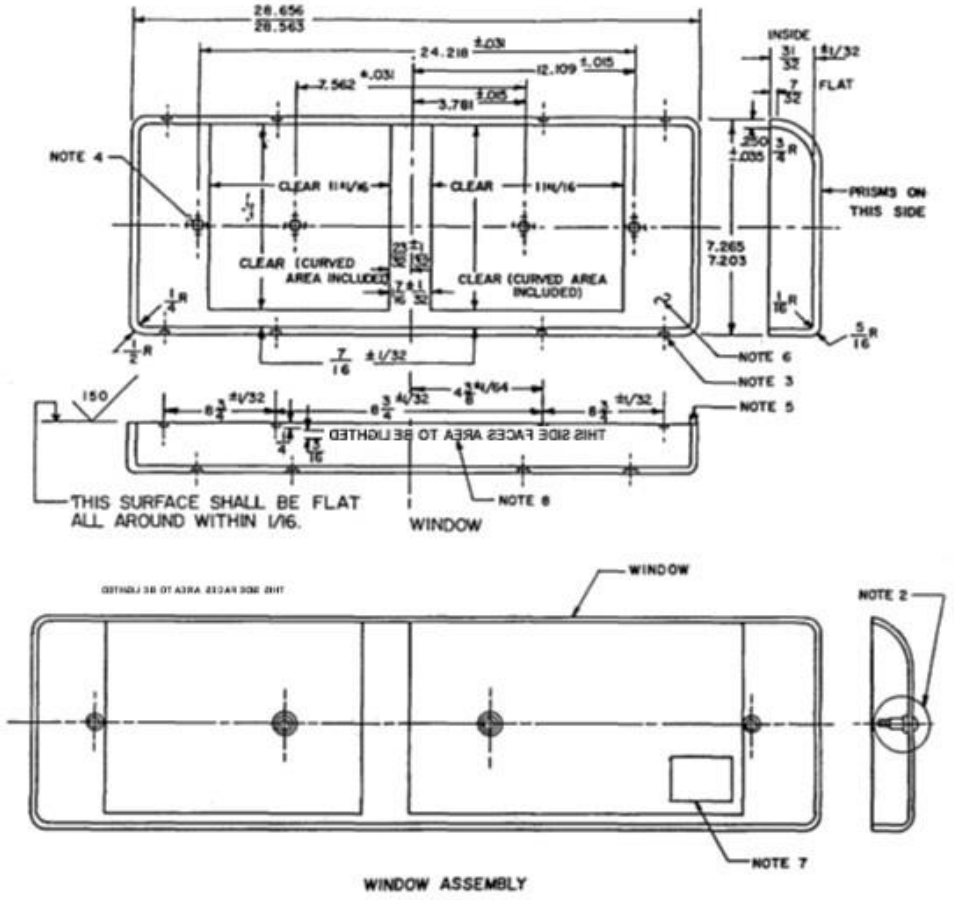


FIGURE 3. Typical section of symbol 145.5.

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WINDOW ASSEMBLY

NOTES:

1. Material for the window shall be ¼-inch-thick clear prismatic acrylic plastic, type CLEAR-2 of MIL-DTL-24191. The prisms shall be on the outside surface of the windows.
2. Four short window-securing screw assemblies shall be installed in accordance with 3.7.4.3 of MIL-DTL-16377.
3. When required, 8 notches (4 on each side) shall be filed as shown in accordance with 3.7.4.4 of MIL-DTL-16377.
4. Drill a 0.191-diameter hole and counterbore opposite side 0.323 diameter, 0.050 deep. Break sharp edges on the inside surface of all holes.
5. Break sharp edges 1/64 of an inch around entire window.
6. The inside surface of the window (except areas indicated as CLEAR) shall be finished white in accordance with 3.8.4.1 of MIL-DTL-16377.
7. For replacement parts only, one type VI modified label shall be applied on the inside surface of curved area of window.
8. The type VI label shall be in accordance with 3.7.4.3 of MIL-DTL-16377, except add "THIS SIDE FACES AREA TO BE LIGHTED".

FIGURE 4. Window assembly, clear and white prismatic – directional.

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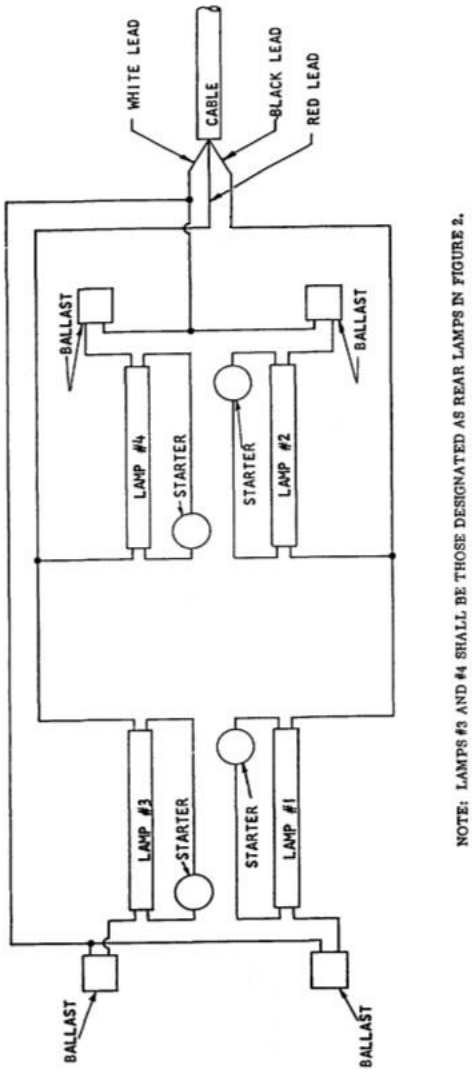
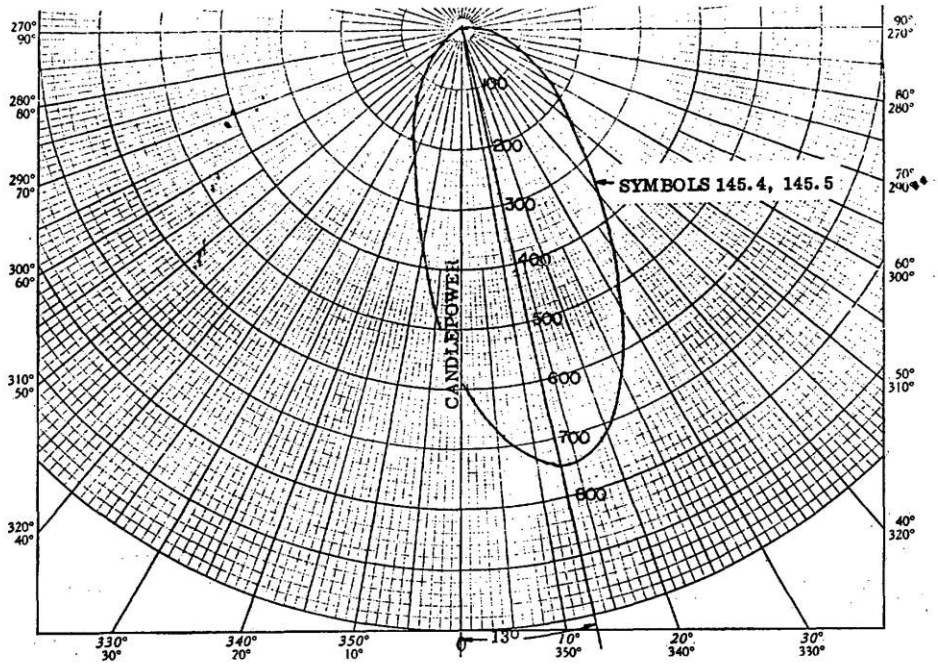


FIGURE 5. Fluorescent wiring diagram.

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*ANGLE	CANDLEPOWER
	SYM 145.5, 145.5
0	740
5	720
10	650
15	560
20	450
25	360
30	290
35	240
40	200
45	160
50	120
55	108
60	80

*ANGLE	CANDLEPOWER
	SYM 145.4, 145.5
65	40
70	20
75	10
80	0
85	0
90	0
95	0
100	0
105	0
110	0
115	0
120	0
125	0

*Measured at the angle of 13° from vertical.

Photometric test data:

Distance: 10 feet

Plane: At right angles to long axis of fixture through geometric center of lamp

Voltage: 115 volts, 60 hertz

Ambient temperature: 76 °C

Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours

Ballast: Use ballast furnished with fixture

Lamp correction: Distribution curves based on 380 lumens per lamp

FIGURE 6. Candlepower distribution curves.

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FLUORESCENT REQUIREMENTS

CLASSIFICATION

Type: I
Class: 2

SYMBOL NUMBERS

145.4: Complete lighting fixture assembly, standard mounting with four 8-watt, cool white fluorescent lamps and clear and white prismatic directional window assembly.

145.5: Complete lighting fixture assembly, flush mounting, with four 8-watt, cool white fluorescent lamps and clear and white prismatic directional window assembly.

DESIGN

Electrical rating:	32 watts, 120 volts, 60 hertz.
Dimension and configurations:	See figures 1, 2, and 3 of this specification sheet and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 6.
Lampholder spacing:	See figure 21 of MIL-DTL-16377.
Wiring diagram:	See figure 5.
Mounting dimensions:	
Symbol 145.4:	See figure 25 of MIL-DTL-16377.
Symbol 145.5:	See figure 26 of MIL-DTL-16377.
Window assembly:	See figure 4.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Cable:	10 feet of type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbol 145.4:	12 pounds, 0 ounces maximum.
Symbol 145.5:	13 pounds, 13 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 1 and 3: (a) Two type I. (b) One type II for symbol 145.5 only. (c) One type IV. The information to be shown on the type IV label shall be: "USE THE FOLLOWING REPAIR PARTS ONLY". List all standard parts shown in table I and window assemblies of table IV by name and stock number only. (d) One type IV, applied as shown on figure 2, bearing the legend: "THIS SIDE OF FIXTURE AND CLEAR, CURVED PORTION OF WINDOW ASSEMBLY SHALL FACE AREA TO BE LIGHTED".
Standard parts:	The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

SOLID-STATE LIGHTING (SSL) REQUIREMENTS:

CLASSIFICATION

Type: III
Class: 2

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SYMBOL NUMBERS

145.4SSL: Complete lighting fixture assembly, standard mounting with LED luminaire and clear and white prismatic directional window assembly.

145.5SSL: Complete lighting fixture assembly, flush mounting, with LED luminaire and clear and white prismatic directional window assembly.

DESIGN

Electrical rating:	20 watts maximum, 120 volts, 60 hertz.
Dimension and configurations:	See figures 1, 2, and 3 of this specification sheet and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 6 of this specification sheet and 3.7.1 of MIL-DTL-16377.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Failsafe circuit:	See 3.6.17 of MIL-DTL-16377.
Wiring:	Wiring shall maintain same control scheme for light output as legacy fluorescent fixtures.
Mounting dimensions:	
Symbol 145.4SSL:	See figure 25 of MIL-DTL-16377.
Symbol 145.5SSL:	See figure 26 of MIL-DTL-16377.
Window assembly:	See figure 4 for overall dimensions.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Cable:	10 feet of type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbol 145.4SSL:	12 pounds, 0 ounces maximum.
Symbol 145.5SSL:	13 pounds, 13 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377: (a) Two type I. (b) One type II. (c) One type IV. The information to be shown on the type IV label shall be: "USE THE FOLLOWING REPAIR PARTS ONLY", followed by a list of available replacement parts and stock numbers, or if unavailable, manufacturer's part numbers.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II for type I and table III for type III
Window assemblies:	See table IV.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377

GENERAL INFORMATION

Assignment of military part numbers and general information is shown in table V.

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TABLE I. Standard parts for type I fixtures.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	---	6240-00-299-5546	Industry Number F8T5/CW
Lamplock	M16377/40-001	6250-00-399-7039	---
Lampholder	UL 542	6250-00-377-8590	---
Startersocket	UL 542	6250-00-295-2738	---
Starter	ANSI C78.180	6250-00-299-5962	Commercial Designation: FS 5
Ballast	M16377/45-001	6250-00-295-2735	---
Shock mount	M16377/41-009 or M16377/41-010	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-DTL-16377
Red filter tube	M16377/42-001	6210-00-399-6297	Not furnished. See 3.7.7 of MIL-DTL-16377

TABLE II. First article and comparison inspection for complete type I fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Noise	3.5.14	4.8.9	---
Continuity of grounding	3.6.13.4	4.8.12	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	---

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TABLE III. First article and comparison inspection for complete type III fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Power interface	3.6.18	4.8.21	---
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Enclosure effectiveness	3.5.11	4.8.14.4	1/ Watertight, unless otherwise specified.
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14.4	1/ Watertight, unless otherwise specified.
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Noise	3.5.14	4.8.9	1/
Continuity of grounding	3.6.13.4	4.8.12	1/
Photometrics	3.7.1	4.8.6	---
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.6.17	4.8.22	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---
Stress relief	3.8.5	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE IV. First article and comparison inspection for window assemblies.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

Military Part Number M16377/10-	Name of Parts	National Stock Number (NSN)
145.4	Complete fixture, symbol 145.4	6210-00-548-0227
145.5	Complete fixture, symbol 145.5	---
145.4SSL	Complete fixture, solid state, symbol 145.4SSL	---
145.5SSL	Complete fixture, solid state, symbol 145.5SSL	---
002	Window assembly, clear and white prismatic, directional	---

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy − SH
(Project 6210-2023-005)

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